

IBM University Engagement Project Submission

“Exploring the Power of Machine Learning with IBM Cloud”

Project Tile – PowerGuard AI: Intelligent Power System Fault Detection & Classification using Machine Learning.

Domain of Project – Power & Energy

Student Name	Recent Passport Photo	Email ID	Mobile number [WhatsApp]
Shravani Prakash Deshmukh		shravanid111@gmail.com	9373888503

Problem Statement -

Problem Statement: Power System Fault Detection and Classification using Machine Learning

Power distribution systems are critical for delivering uninterrupted electricity, but they are vulnerable to faults such as Transformer Failure, Line Breakage, and Overheating. Delayed or inaccurate fault identification can lead to equipment damage, power outages, increased maintenance costs, and reduced grid reliability.

Traditional fault detection methods often require manual analysis and may not provide rapid diagnosis. Therefore, there is a need for an intelligent system that can automatically analyze electrical parameters and accurately classify different types of power system faults.

This project aims to develop PowerGuard AI, a Machine Learning-based fault detection and classification system that uses electrical measurements such as voltage, current, power load, temperature, wind speed, weather conditions, maintenance status, component health, fault duration, and downtime to predict the type of fault. The system provides a user-friendly web interface for prediction and an analytics dashboard for data visualization, supporting faster fault diagnosis and improved maintenance planning.

Proposed Solution -

We propose **PowerGuard AI**, an intelligent Machine Learning-based system for detecting and classifying faults in power distribution systems. The solution uses historical electrical measurement data, including voltage, current, power load, temperature, wind speed, weather conditions, maintenance status, component health, fault duration, and downtime to predict the type of power system fault.

The dataset is preprocessed by handling categorical features through label encoding and then used to train a Random Forest Classifier. The trained model is integrated into a Flask-based web application, allowing users to enter electrical parameters and receive instant fault predictions. An interactive Analytics Dashboard is also provided to visualize dataset statistics and fault distributions.

The project utilizes IBM Cloud Lite and IBM watsonx.ai as the cloud-based project workspace, with IBM Cloud Object Storage managing project assets. This solution enables faster fault identification, supports predictive maintenance, and provides a scalable foundation for future cloud deployment and real-time monitoring.

Technology Used

- **IBM Cloud** – IBM Cloud Lite provides the cloud platform for creating and managing the AI project. IBM watsonx.ai is used as the project workspace, IBM Cloud Object Storage manages the project assets and dataset, and IBM IAM provides secure authentication through API keys. These services support project organization and provide a foundation for future cloud deployment.
- ML Algorithm/ Model Used: Random Forest Classifier
- Accuracy Obtained: 29.4%
- Enhancements and Hyperparameter tuning:

Performed data preprocessing using Label Encoding for categorical features.

Optimized the Random Forest model using GridSearchCV with the following best parameters:

n_estimators = 200

max_depth = 20

min_samples_split = 2

Training Screenshots-

```
===== DATASET INFO =====
```

```
Fault Type
```

```
Transformer Failure      171
```

```
Overheating              171
```

```
Line Breakage            164
```

```
Name: count, dtype: int64
```

```
Training Random Forest...
```

```
=====
```

```
Best Parameters
```

```
{'max_depth': 20, 'min_samples_split': 2, 'n_estimators': 200}
```

```
=====
```

```
Accuracy : 29.41%
```

```
=====
```

```
=====
```

```
Best Parameters
```

```
{'max_depth': 20, 'min_samples_split': 2, 'n_estimators': 200}
```

```
=====
```

```
Accuracy : 29.41%
```

```
=====
```

```
Best Parameters
```

```
{'max_depth': 20, 'min_samples_split': 2, 'n_estimators': 200}
```

```
=====
```

```
Accuracy : 29.41%
```

```
=====
```

```
{'max_depth': 20, 'min_samples_split': 2, 'n_estimators': 200}
```

```
=====
```

```
Accuracy : 29.41%
```

```
=====
```

```
=====
```

```
Accuracy : 29.41%
```

```
=====
```

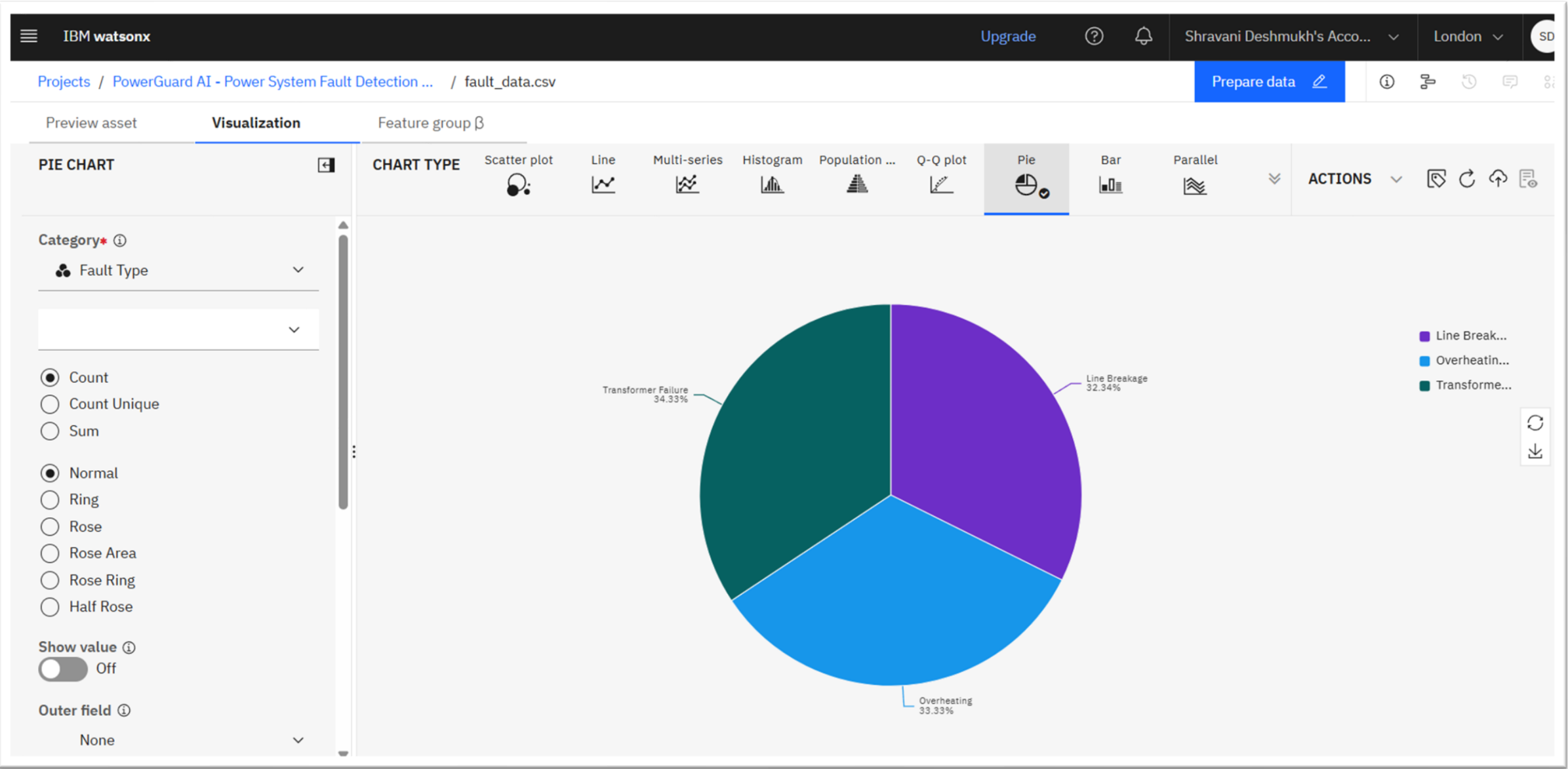
```
Accuracy : 29.41%
```

```
=====
```

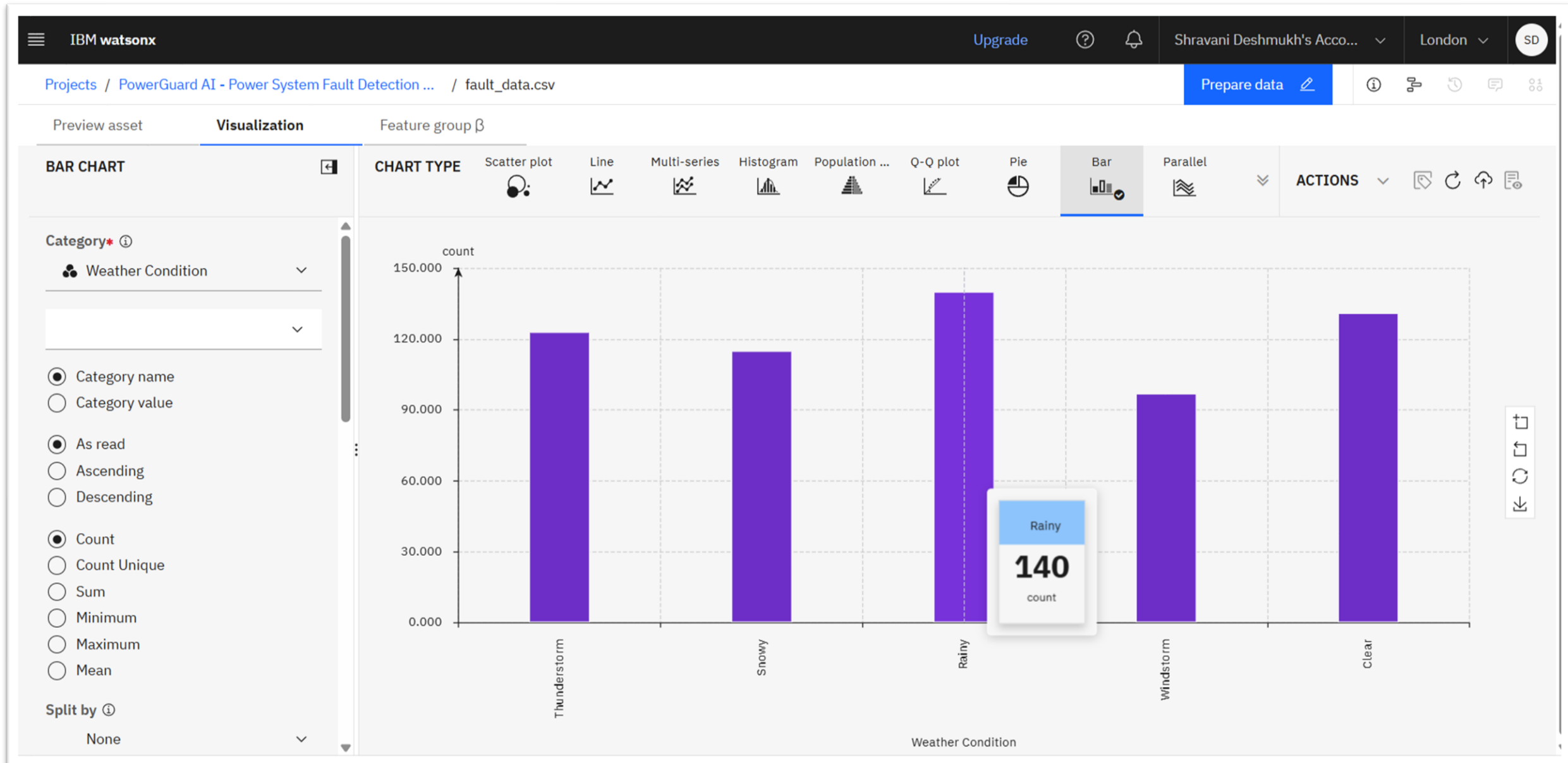
Best Model Screenshot:

=====				
	precision	recall	f1-score	support
=====				
	precision	recall	f1-score	support
0	0.27	0.24	0.25	33
	precision	recall	f1-score	support
0	0.27	0.24	0.25	33
1	0.33	0.37	0.35	35
0	0.27	0.24	0.25	33
1	0.33	0.37	0.35	35
0	0.27	0.24	0.25	33
1	0.33	0.37	0.35	35
1	0.33	0.37	0.35	35
2	0.27	0.26	0.27	34
accuracy			0.29	102
macro avg			0.29	102
weighted avg			0.29	102
Model Saved Successfully!				

Testing Screenshot:



Testing Screenshot:



Novelty and Uniqueness

- Predicts multiple power system fault types using Machine Learning.
- Interactive web application developed using Flask.
- Analytics dashboard provides visual representation of dataset statistics.
- Integrates IBM Cloud Lite and IBM watsonx.ai project workspace.
- Supports early fault detection to improve maintenance planning.
- Simple, scalable, and suitable for future smart grid applications.

Git Hub Link

Please test and Paste the working Github URL –

<https://github.com/Shravanid31/PowerGuard-AI.git>

Future Scope

- Integrate IoT sensors to collect real-time power system data for continuous fault monitoring.
- Improve prediction accuracy by using larger and more diverse real-world datasets.
- Explore advanced machine learning and deep learning models such as XGBoost and Neural Networks.
- Implement real-time alerts and notifications for maintenance teams when a fault is detected.
- Develop a mobile application for remote monitoring and instant fault prediction. Integrate weather and environmental data APIs to further enhance fault prediction performance.
- Extend the system to support predictive maintenance, helping utilities reduce downtime and maintenance costs.

Certificates Earned:

In recognition of the commitment to achieve
professional excellence



Shravani Deshmukh

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence



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Certificates Earned:

IBM **SkillsBuild**

Completion Certificate



This certificate is presented to

Shravani Deshmukh

for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 19 Jun 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.